

FHIR + AI Hackathon - Session 2 Orientation

Session 2: Understanding Tool Use - The Agentic Pattern

Duration: 1 hour

Recap: Session 1

You orchestrated the pipeline:

You made every decision about what to do next.

Session 2: The LLM Decides

Today's question:

You provide:

The LLM decides:

What is Tool Use?

Also called:

How it works:

Key point: The LLM never executes code. It requests function calls. Your code runs them.

Tool Use Flow

User Question

The Three FHIR Tools

Tool 1: search_conditions

Tool 2: get_patient

Tool 3: search_observations

Example: Tool Schema

json

The LLM reads this and knows:

Clinical Codes (Same as Session 1)

| Code | System | Meaning | Reference |

Remember:

The LLM will learn these from the tool descriptions!

System Prompt: Guiding the Agent

You are a clinical data assistant with access to a FHIR server.

When asked a clinical question, use the available tools to

1. Search for relevant conditions

Rules:

The system prompt teaches the LLM how to think about the task.

What You'll Observe Today

The Agent Loop in Action

1. First decision: LLM calls `search_conditions(code='44054006')`
2. Second decision: LLM calls `get_patient(...)` for each patient
3. Third decision: LLM calls `search_observations(patient_id=..., loinc_code='4548-4')`
4. Final step: LLM synthesizes answer

You'll see every tool call in the trace!

Prediction Exercise

Before running the agent, predict:

1. What will the agent's first tool call be?

Compare your predictions to what actually happens.

Common Patterns You'll See

Sequential Calls

Error Handling

Efficiency

Session 1 vs Session 2 Comparison

| Aspect | Session 1 | Session 2 |

Why This Matters

Traditional Approach (Session 1)

Agent Approach (Session 2)

Failure Modes to Watch For

- 1. Hallucination**
- 2. Wrong Tool Choice**
- 3. Premature Termination**
- 4. Over-Querying**

Safeguards in the Notebook

System Prompt

Tool Descriptions

Max Steps Limit

Your Code Executes Tools

The LLM plans, but your code maintains control!

What to Expect

[OK] Success:

[!] Common observations:

Next Session Preview

Session 3: Open-Ended Exploration

Today:

Session 3:

Tips for Success

1. Read the tool call trace carefully
2. Check the final answer against tool results
3. Think about the system prompt

Questions?

Before you start:

Let's watch the agent work!

Resources

- Anthropic Tool Use Docs: <https://docs.anthropic.com/claude/docs/tool-use>

Key Codes: